

Pipeline™ Flex Embolization Device with Shield Technology™

Shielded by science

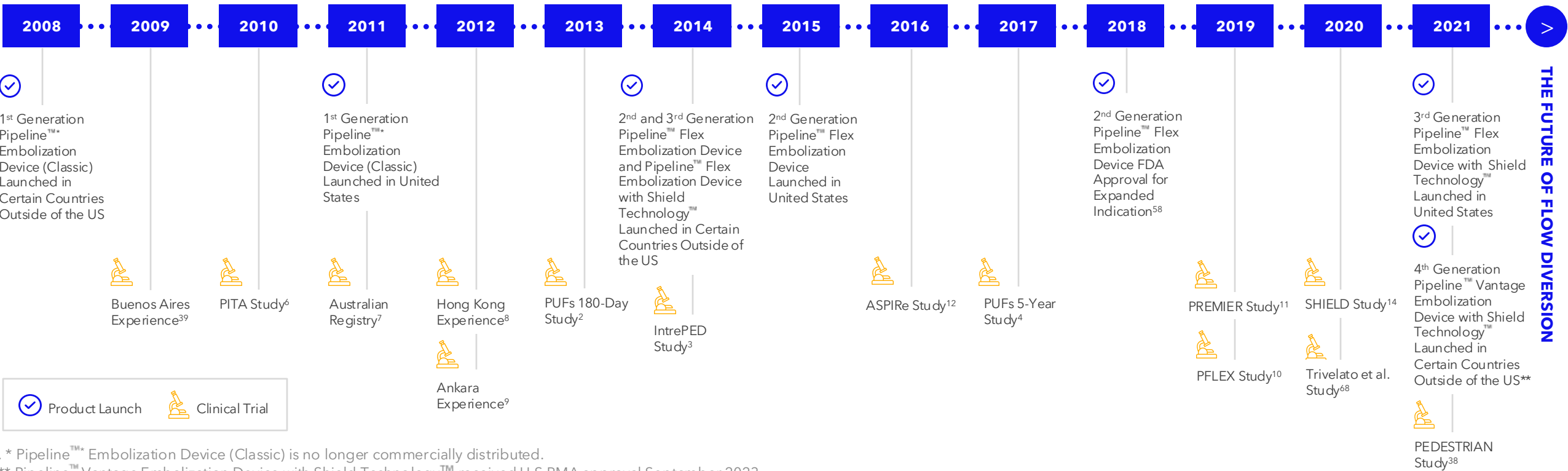


Medtronic Neurovascular has been leading the way in flow diversion for over 10 years

Our lasting commitment to clinically proven safety and efficacy is demonstrated in our legacy of robust clinical trials.

Through this commitment, we are driven towards always advancing flow diversion therapy of intracranial aneurysms.

Timeline



. * Pipeline™* Embolization Device (Classic) is no longer commercially distributed.
** Pipeline™ Vantage Embolization Device with Shield Technology™ received U.S PMA approval September 2023.



Timeline

Science and
Technology

Indication

Pipeline™ Flex with Shield Technology™

is designed for the treatment of certain intracranial aneurysms.⁵





Timeline

Science and
Technology

Indication

Pipeline™ Flex with Shield Technology™ is indicated for:⁵

01.

The endovascular treatment of adults (22 years of age or older) with large or giant wide-necked intracranial aneurysms (IAs) in the internal carotid artery from the petrous to the superior hypophyseal segments.

[For supporting data, see PUFs Study^{2,4}](#)

02.

Use in the internal carotid artery up to the terminus for the endovascular treatment of adults (22 years of age or older) with small and medium wide-necked (neck width ≥ 4 mm or dome-to-neck ratio < 2) saccular or fusiform intracranial aneurysm (IAs) arising from a parent vessel with a diameter ≥ 2.0 mm and ≤ 5.0 mm.

[For supporting data, see Premier Study¹¹](#)

Contraindications of Pipeline™ Flex with Shield Technology™:5

Timeline
Science and Technology
Indication
◀ Back Next ▶

1.

Patients with active bacterial infection.
2.

Patients in whom dual antiplatelet therapy (aspirin and clopidogrel) is contraindicated.
3.

Patients who have not received dual antiplatelet agents prior to the procedure.
4.

Patients in whom a pre-existing stent is in place in the parent artery at the target aneurysm location.
5.

Patients in whom the parent vessel size does not fall within the indicated range.

MR Conditional⁵

Non-clinical testing has demonstrated that the Pipeline™ Flex Embolization Device with Shield Technology™ is MR Conditional for single and overlapping stents up to 70 mm in length. A patient with this device can be scanned safely in an MR system immediately after placement under the following conditions:

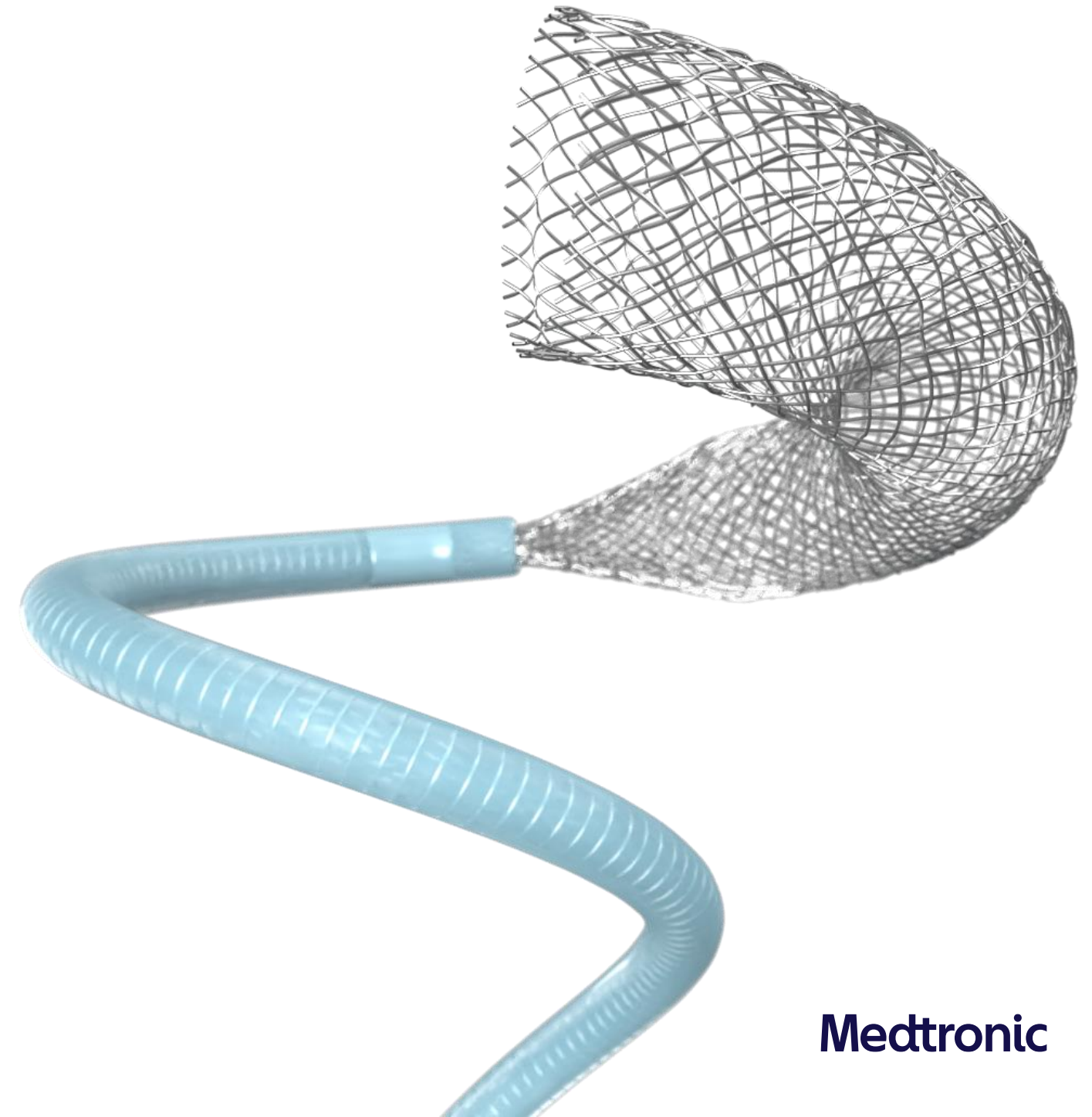
Parameter	Condition of Use / Information
Static Magnetic Field Strength (B ₀) [T]	1.5T, 3T
Type of Nuclei	Hydrogen Proton
Static Magnetic Field (B ₀) Orientation	Horizontal, Cylindrical bore
Maximum Spatial Field Gradient (SFG) [T/m] and [gauss/cm]	30 T/m (3000 gauss/cm)
RF Polarization	Circularly Polarized (CP) (i.e., quadrature drive)
RF Transmit Coil	Integrated Whole Body Transmit RF Coil Detachable Extremity Transmit / Receive RF coil
RF Receive Coil	Any receive-only RF coil may be used
MR System (RF) Operating Modes or Constraints	Normal Operating Mode
Whole Body Averaged SAR [W/kg]	≤ 2 W/kg
Head SAR [W/kg]	≤ 3.2 W/kg

Parameter	Condition of Use / Information
Scan Duration	60 minutes of continuous radiofrequency (RF) (a sequence or back to back series/scan without breaks) followed by a wait time of 5 minutes if this limit is reached.
Anatomy at Isocenter	Any anatomic location at isocenter is acceptable
Patient position in scanner	Any patient position is acceptable
MR Image Artifact	In non-clinical testing, the image artifact caused by the Pipeline™ Flex Embolization Device with Shield Technology™ extends approximately 11.0 mm from this implant when imaged using a T1-weighted spin echo pulse sequence and a 3-Tesla MR system. Local field artifact from the Pipeline™ Flex Embolization Device with Shield Technology™ may decrease the accuracy of the MR angiogram in assessing vessel luminal patency. MR image quality may be compromised if the area is in the exact same area or relatively close to the position of the Pipeline™ Flex Embolization Device with Shield Technology™. Therefore, it may be necessary to optimize MR imaging parameters for the presence of this metallic implant.

CAUTION: The MRI safety when using a detachable head transmit/receive RF coil to image a patient with the Pipeline™ Flex Embolization Device with Shield Technology™ has not been evaluated. Use of the integrated whole body transmit coil with a receive-only head coil is acceptable.

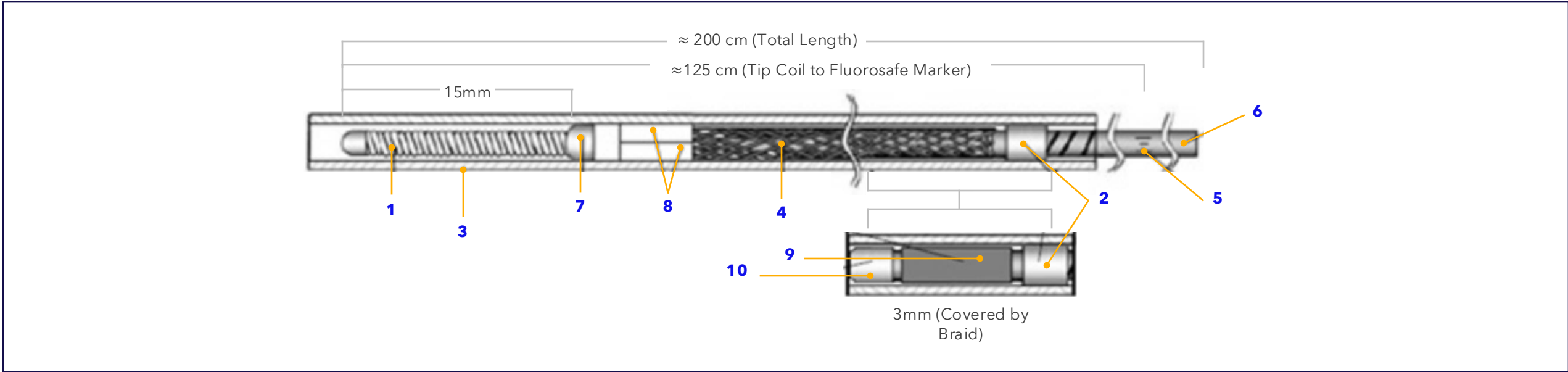


Overall Design



THE PIPELINE™ FLEX WITH SHIELD TECHNOLOGY™5

implant is mounted on a ~200 cm long micro-guidewire and compressed inside an introducer sheath



- 1

Tip Coil
- 2

Proximal Bumper
- 3

Introducer Sheath
- 4

Braid
- 5

Fluorosafe Marker
- 6

Delivery Wire
- 7

Distal Maker
- 8

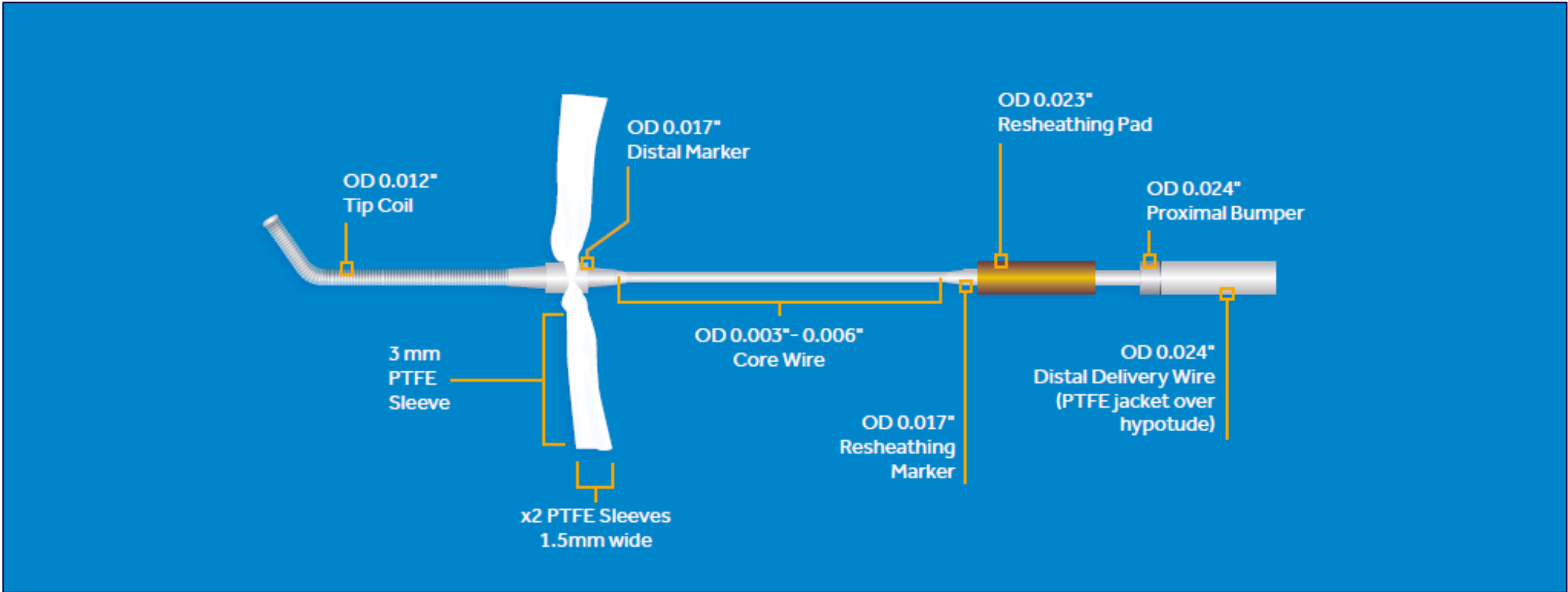
PTFE Sleeves
- 9

Resheathing Pad
- 10

Resheathing Maker



The Pipeline™ Flex with Shield Technology™ 18-27



Overall Design

Braid Design

Delivery System Design

Shield Technology™

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Overall Design



A single Pipeline™ Flex with Shield Technology™ unit contains:⁵

- ▶ One Pipeline™ Flex with Shield Technology™ device
- ▶ One introducer sheath
- ▶ One flexible delivery wire
- ▶ One torquer

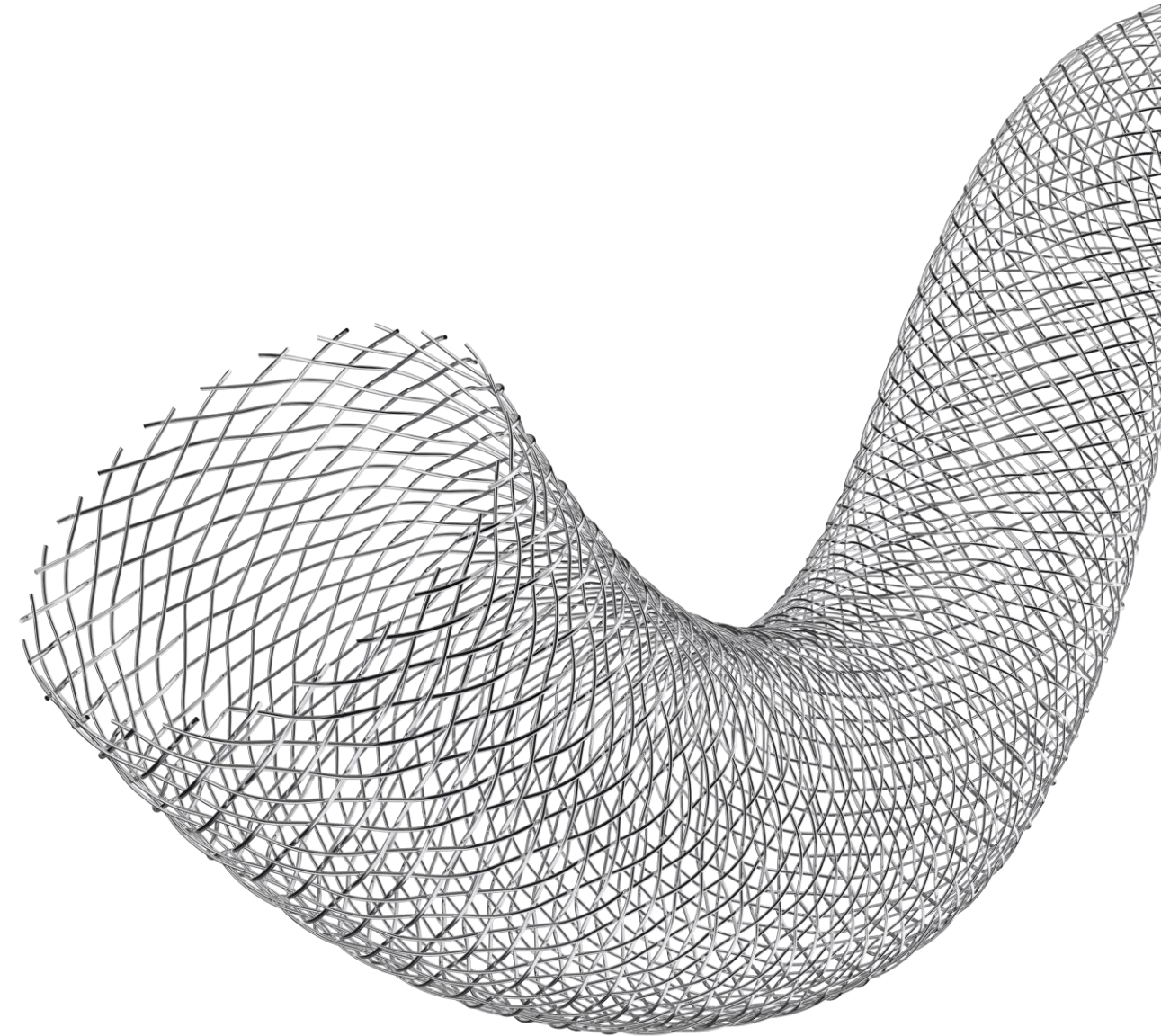


Compatibility:⁵

The Pipeline™ Flex with Shield Technology™ device is designed to be delivered only through a compatible **micro catheter of 0.027 inch** (0.69 mm) inside diameter at least 135 cm in length.

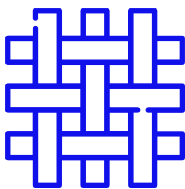


Braid Design



The Pipeline™ Flex with Shield Technology™ implant is a braided, multi-alloy, mesh cylinder woven from platinum-tungsten and cobalt-chromium alloy wires⁵

Overall Design



48 wires interwoven in standard pattern for all braid diameters³⁴

Braid Design

Delivery System Design

Shield Technology™

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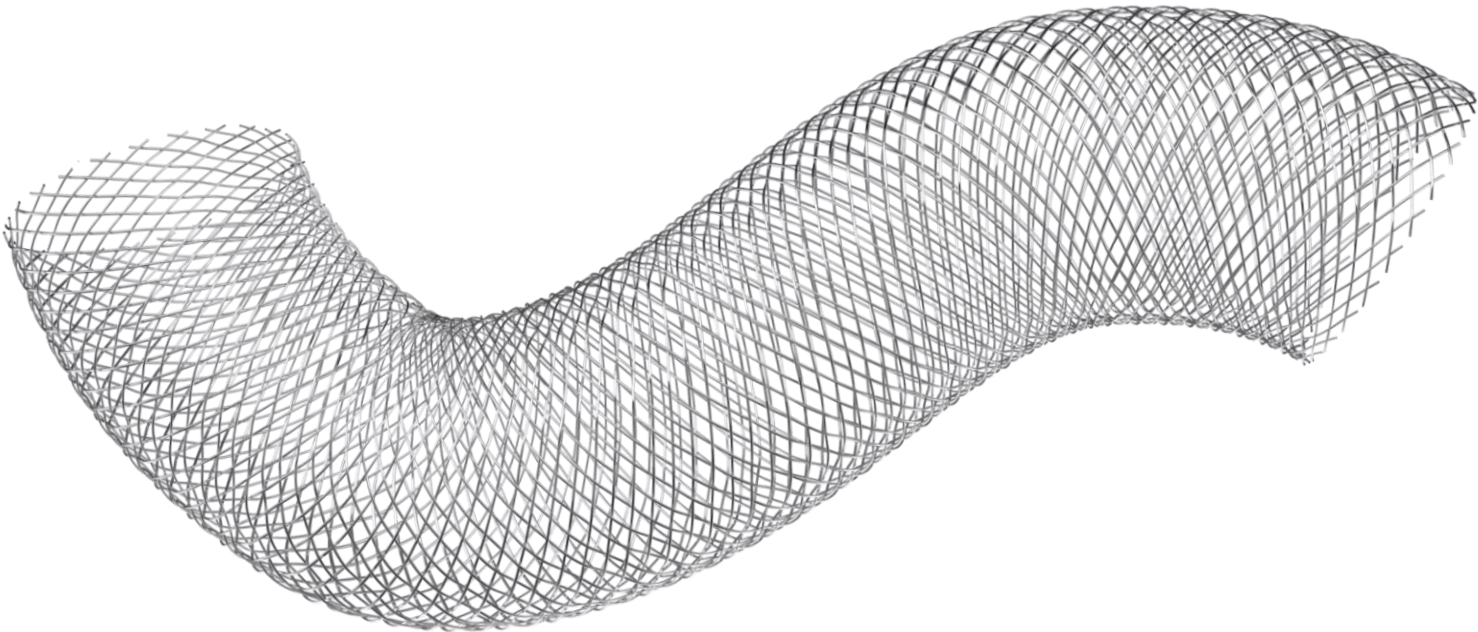
Balance visibility and radial strength^{33,34*}

- 1. 75% cobalt-chromium alloy
- 2. 25% platinum-tungsten alloy

*Data is derived from referenced bench studies and may not be representative of clinical performance



Braid Sizing⁵



Size Availability⁵

2.50-5.00mm
Diameters (D)

10-35mm
Lengths (L)

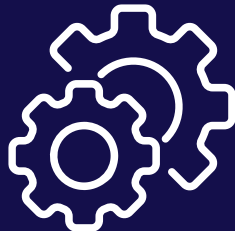
NOTE: Not all lengths are available for each diameter.



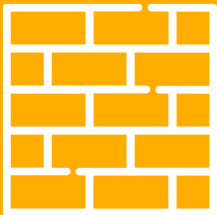
Three key variables of the braid:



Pore Density



Metal Coverage



Wall Apposition
and Thickness

Overall Design

Braid Design

Delivery System
Design

Shield
Technology™

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Pore Density

Pipeline™ Flex with Shield Technology™

Overall Design

Braid Design

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Porosity

The fraction of metal-free area per total stent surface area coverage. Porosity is closely related to the pore density of the braid, the number of pores per unit area along the device-neck interface.¹⁶

The optimal pore density provides a balance between intra-aneurysmal flow, the volumetric flow into the aneurysm sac, and device flexibility and maneuverability.³⁵

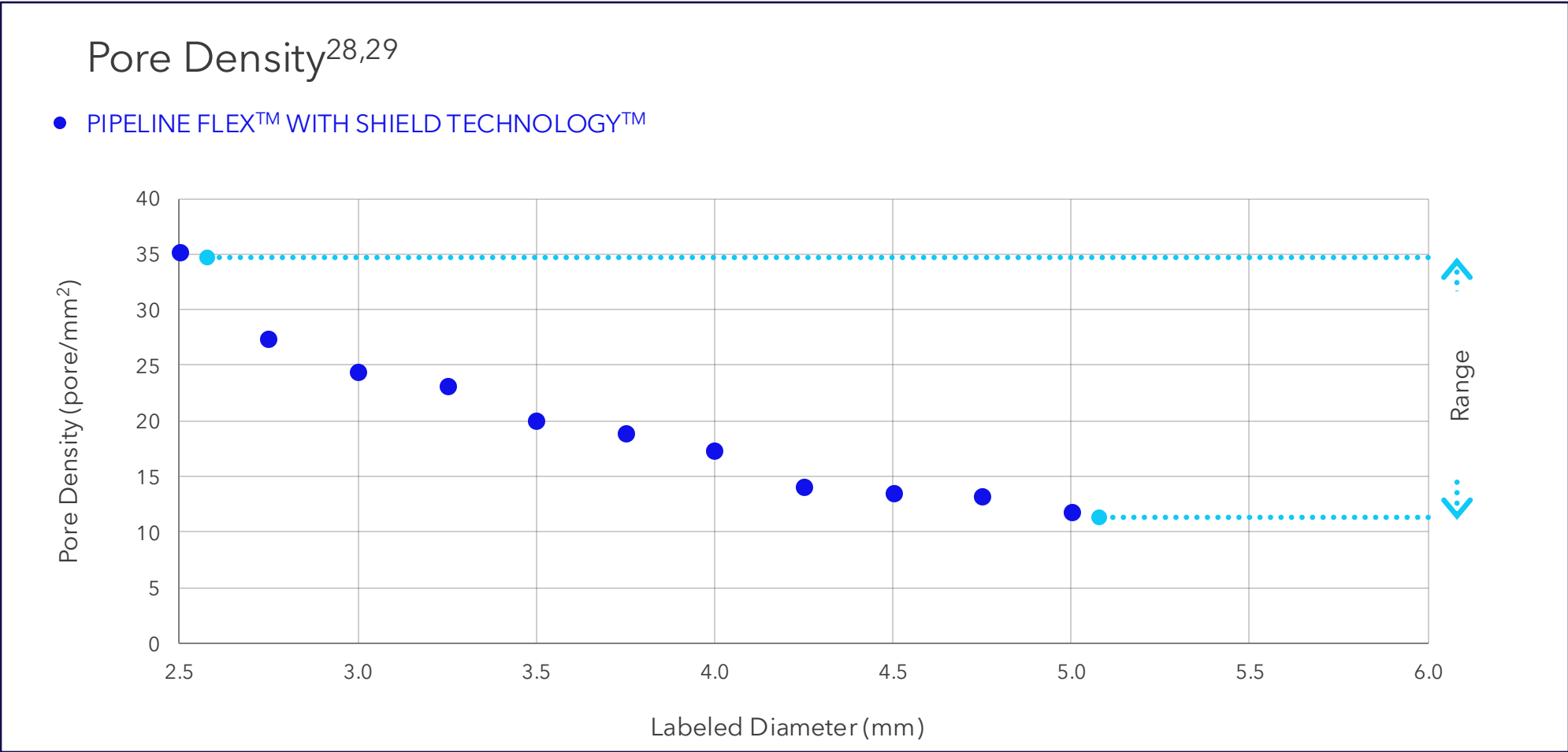


Porosity^{28, 29}
~68%-73%

*Pipeline™ Flex embolization device with Shield Technology has the same porosity as Pipeline™ Flex embolization device and Pipeline™ (Classic) embolization device.

Pore Density

Pipeline™ Flex with Shield Technology™



Pore Density^{28,29}
~12 to 35 pore/mm²

*Pipeline™ Flex embolization device with Shield Technology has the same porosity as Pipeline™ Flex embolization device and Pipeline™ (Classic) embolization device.



Metal Coverage

Pipeline™ Flex with Shield Technology™

Overall Design

Pipeline™ Flex with Shield Technology’s metal coverage is backed by over a decade of clinical evidence^{36*}

Braid Design

The Effects of Metal Coverage:

Delivery System Design

Too Much:
Risk of occluding the aneurysm neck or branch vessel³⁰

Shield Technology™

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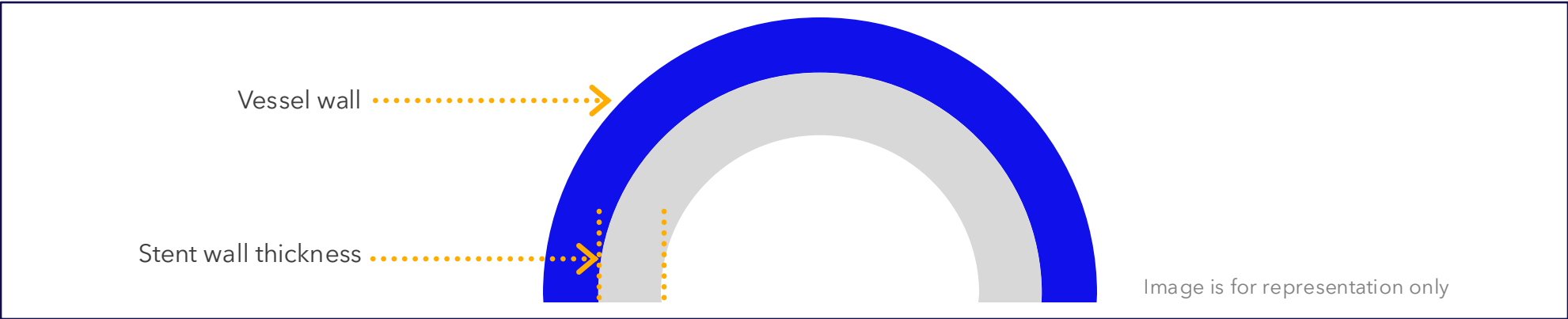


Metal Coverage²⁸
~27%-32%



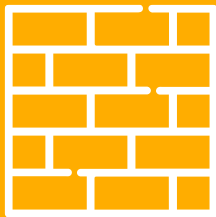
Wall Apposition and Thickness

Pipeline™ Flex with Shield Technology™



01 Pipeline™ Flex with Shield Technology™ provides a low wall profile design to allow blood flow in the native vessel to remain as unimpeded as possible while maintaining wall apposition.³²

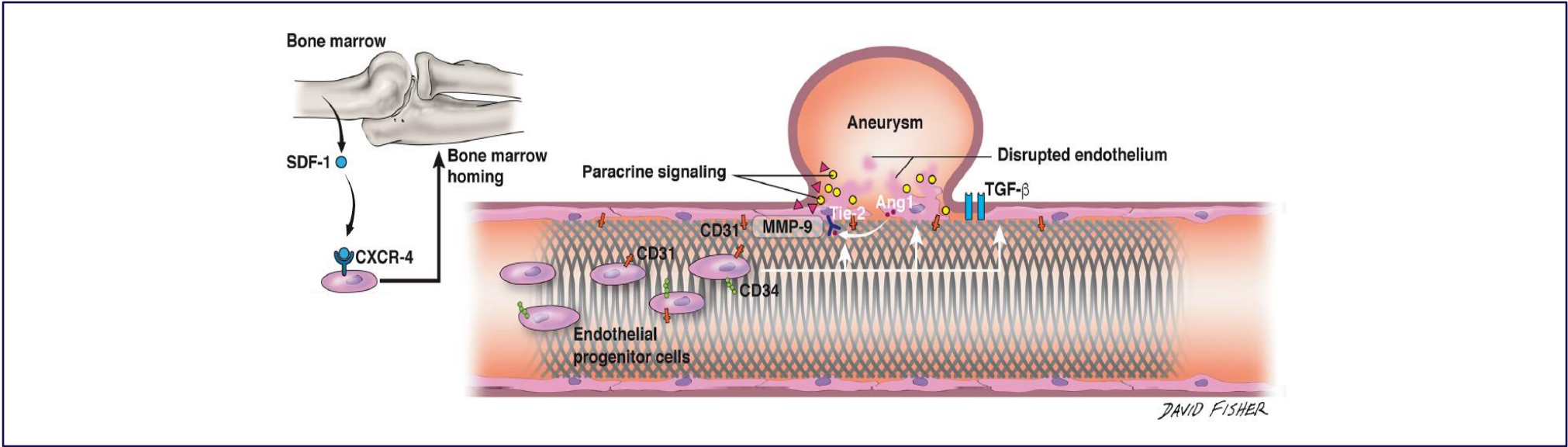
02 It is believed that direct flow diverter contact with the wall is necessary to provide a scaffold for contiguous endothelial cell growth from the parent vessel, and thus, strut thickness may represent an important engineering variable for optimization.³¹



Wall
Thickness³²
~0.0023in



Endothelialization following flow diversion treatment¹⁶



Cellular and molecular mechanisms underpinning device

Depicted are the contributing roles of local endothelial cells, activated by paracrine factors at site flow diverter deployment, and circulating endothelial progenitor cells origination from the bone marrow.

Artist: David Fisher, Department of Neurosurgery, University of Alabama-Birmingham. Published with permission.

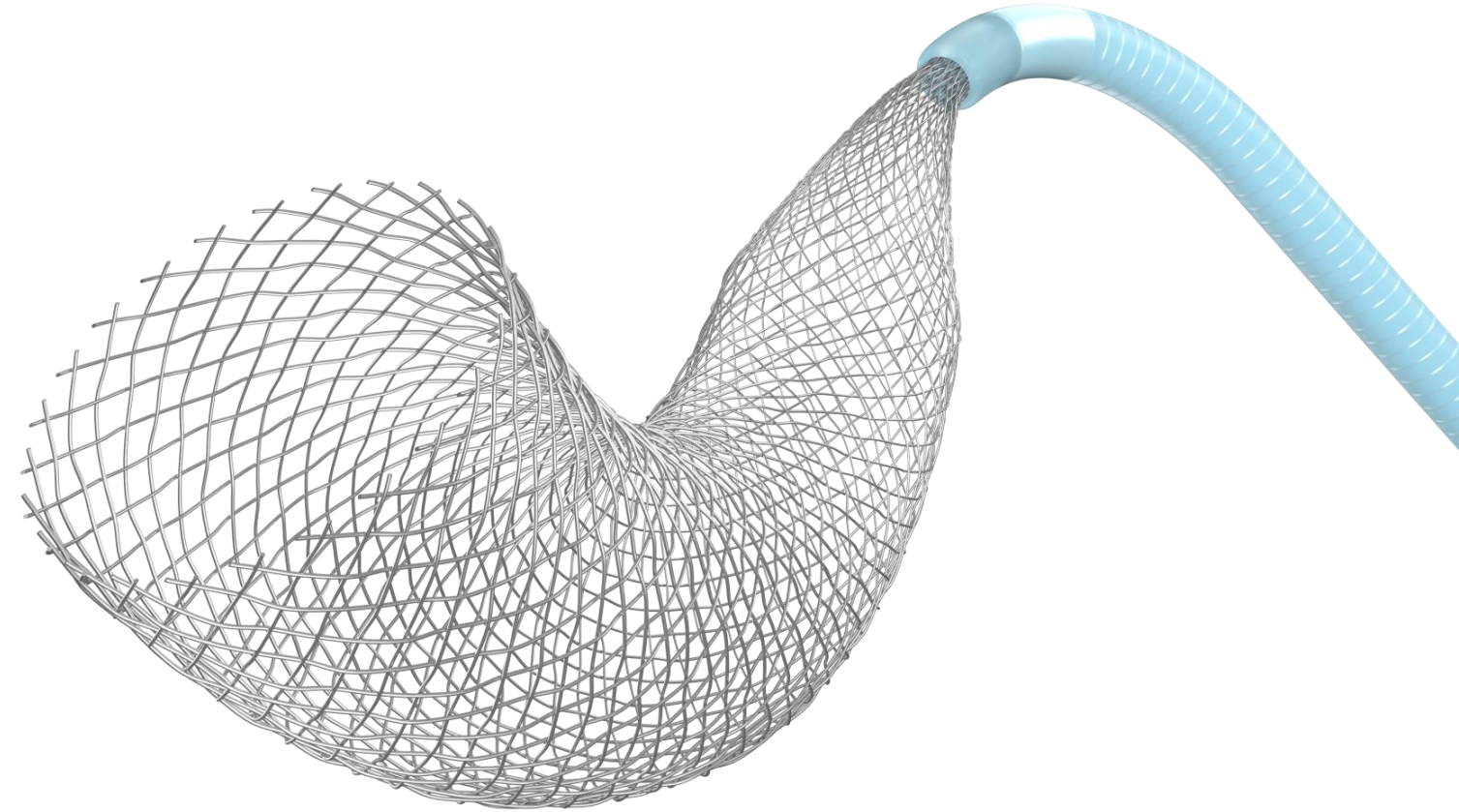


For more information on the cellular and molecular mechanisms contributing to flow diversion, see:

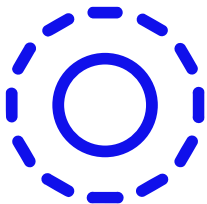
Ravindran, K. et al.
Mechanism of Action and Biology of Flow Diverters in the Treatment of Intracranial Aneurysms. Neurosurgery 86: S13-S19, 2020



Delivery System Design



Overall Design



Pusher Wire Design^{25,33*}
Proximal support and pushability

Braid Design

Delivery System Design



Resheathing Mechanism^{5,33*}
Resheathable to allow for repositioning and redeployment

Shield Technology™



Distal End^{5,33*}
Distal navigability to maneuver through tortuous anatomy

*Data is derived from referenced bench studies and may not be representative of clinical performance



Pusher Wire Design

Overall Design

Braid Design

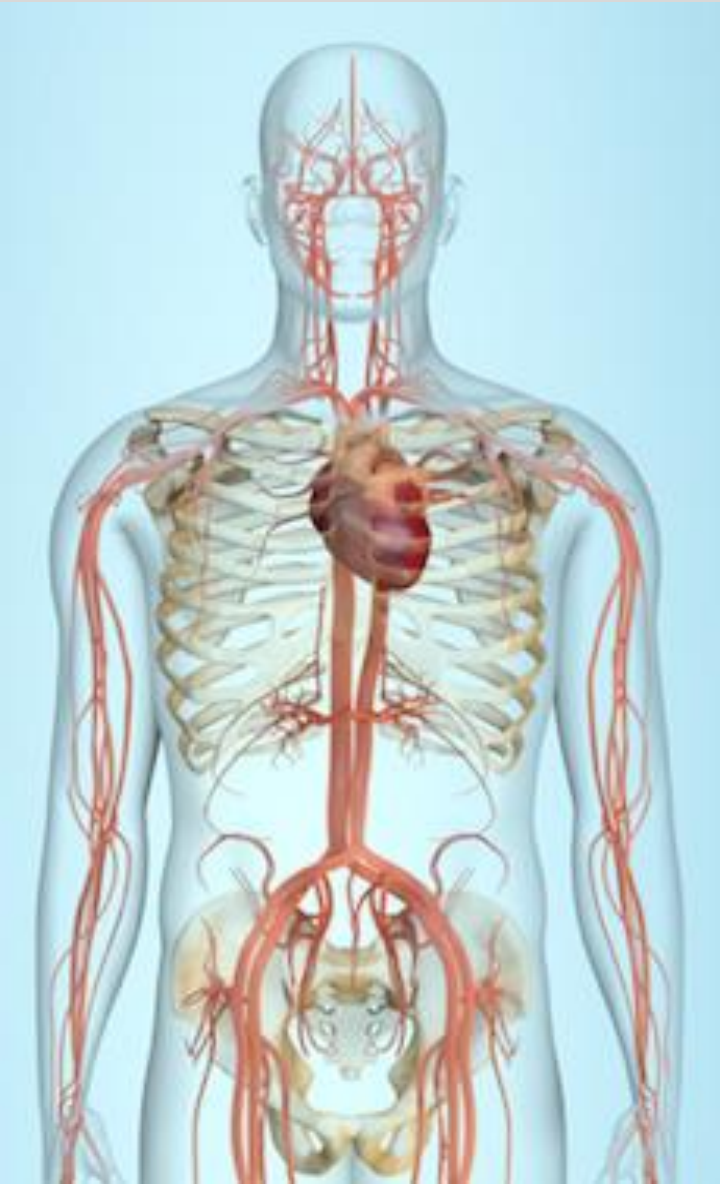
Delivery System Design

Shield Technology™

Four anatomically designed flex-zones are laser cut along a stainless steel hypotube²⁵

Designed to provide proximal support and pushability^{25,33*}

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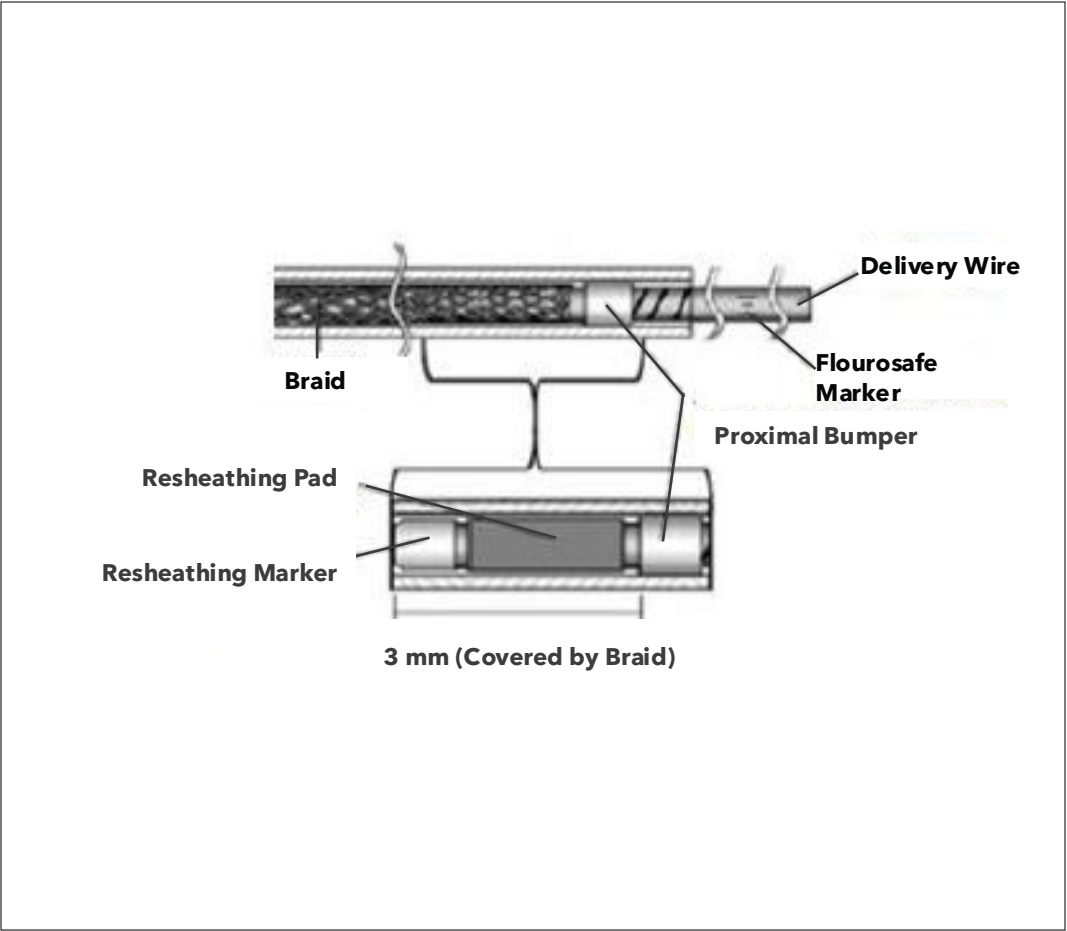
*Data is derived from referenced bench studies and may not be representative of clinical performance



Resheathing Mechanism⁵

Able to capture full system and redeploy up to two times for repositioning

- 01
- The proximal bumper and resheathing pad allows the user to push the Pipeline™ Flex with Shield Technology™ device out of the micro catheter when the delivery system is advanced.
- 02
- The resheathing pad allows the user to resheath the Pipeline™ Flex with Shield Technology™ device back into the micro catheter.
- 03
- The resheathing marker provides the user fluoroscopic visualization for the limit of resheathing the Pipeline™ Flex with Shield Technology™ device.



Overall Design

Braid Design

Delivery System Design

Shield Technology™

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Distal End^{5,13}



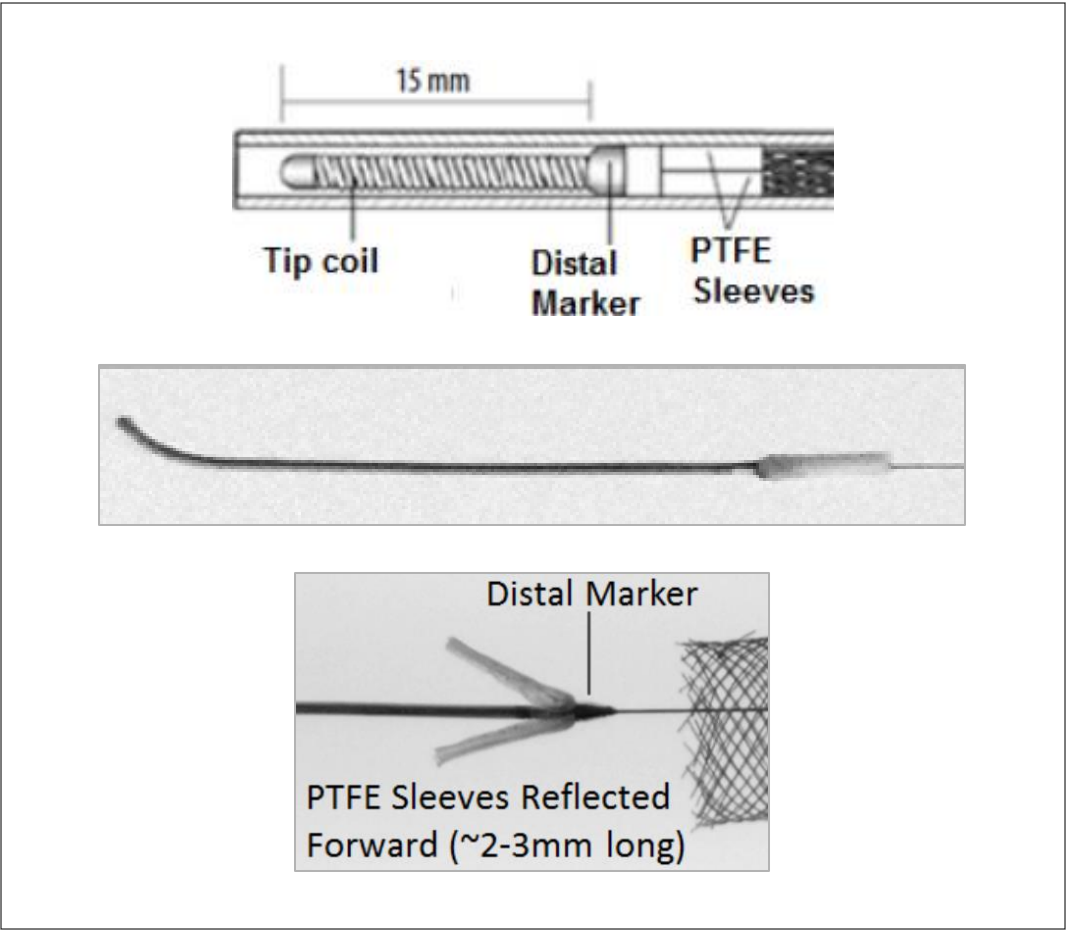
The protective sleeves provide protection to the distal portion of the braid while advancing through the micro catheter



Platinum tip coil provides visibility



Soft, angled tip coil^{18,33*}



*Data is derived from referenced bench studies and may not be representative of clinical performance



Shield Technology™





Overall Design

Braid Design

Delivery System Design

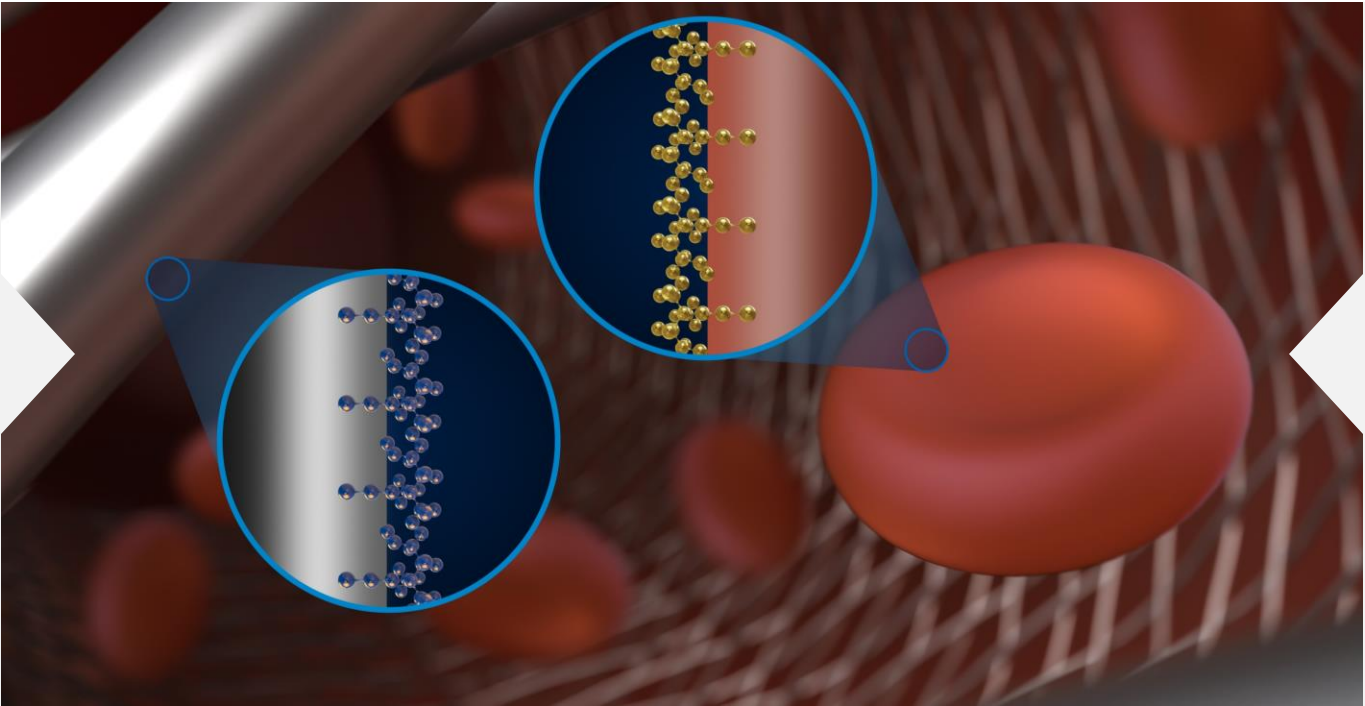
Shield Technology™

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Through covalently bonding phosphorylcholine to the surface of the implant, Shield Technology™ is a surface modification on the Pipeline™ Flex embolization device



Synthetic phosphorylcholine polymer is covalently bonded to the braid, and designed to provide a biomimetic surface modification⁴⁴



Phosphorylcholine is present in the outer membrane of red blood cells³⁷



Overall Design

Braid Design

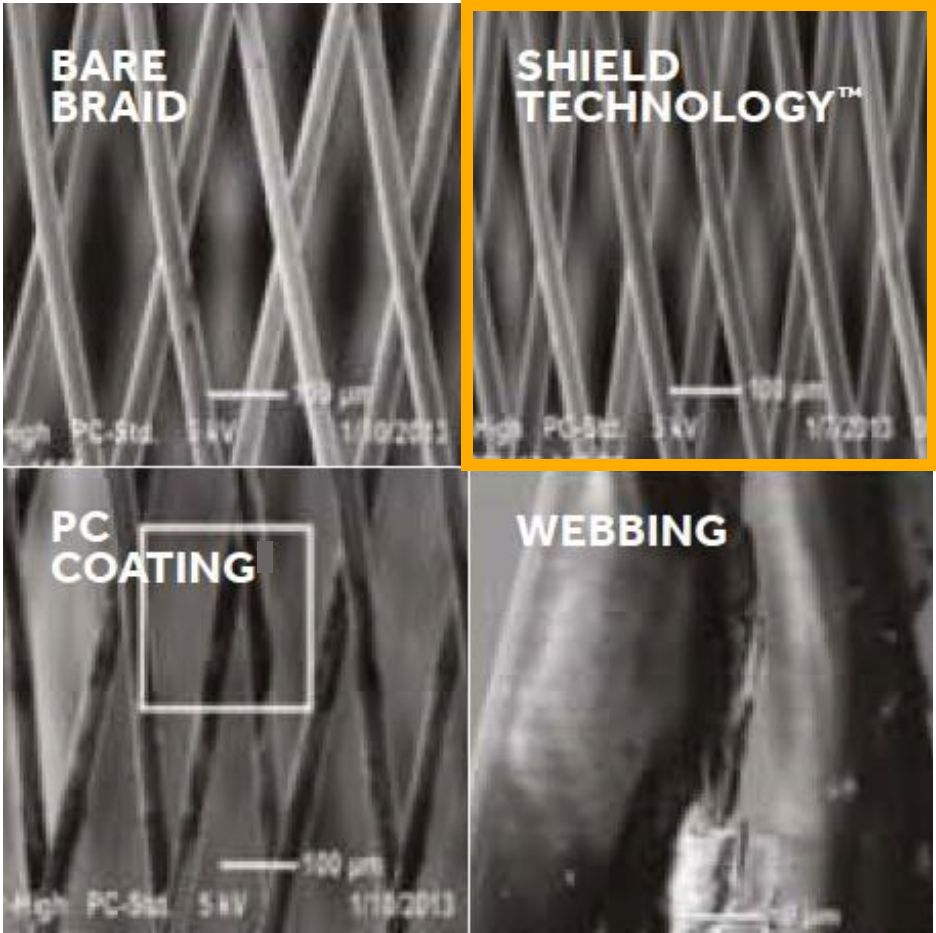
Delivery System Design

Shield Technology™

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Traditional modifications are adhered to the substrate by physical encapsulation through a process of dipping, spraying or brushing while Shield Technology™ is a **surface modification** adhered to the implant by covalent bonding through a chemical reaction process.^{15,17,*}

The thickness of Shield Technology™ is less than ~3 nanometers (~0.01% of one braid wire)⁵²



*Data is derived from referenced pre-clinical studies and bench testing and may not be representative of clinical performance.

Shield Technology™ is Backed by Science

Overall Design
Braid Design
Delivery System Design
Shield Technology™

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01

THROMBOGRAM ANALYSIS^{17*}

Published: Girdhar G, et al. In-vitro thrombogenicity assessment of flow diversion and aneurysm bridging devices. Journal of Thrombosis and Thrombolysis, 2015.

02

HUMAN BLOOD LOOP MODEL^{49*}

Girdhar, G et al. Thrombogenicity assessment of Pipeline Flex, Pipeline Shield, and FRED flow diverters in an in vitro human blood physiological flow loop model, Journal of Biomedical Materials Research 2018.

03

EX VIVO PRIMATE SHUNT MODEL^{48*}

Key Outcome:
Lower levels of platelet deposition levels with Pipeline Flex Embolization Devices with Shield Technology™ were demonstrated with DAPT

Published: Hagen, MW, et al. Thrombogenicity of flow diverters in an ex vivo shunt model: effect of phosphorylcholine surface modification, JINIS, 2017.

04

RABBIT ELASTASE ANEURYSM MODEL^{47,46*}

Key Outcome:
Significantly less clot formation at side branches, better wall apposition, less neointimal hyperplasia and similar aneurysm occlusion

Published: Marosfoi M, et al. Acute thrombus formation on phosphorylcholine surface modified flow diverters, JINIS, 2017.
Caroff J, et al. Phosphorylcholine surface modified flow diverter associated with reduced intimal hyperplasia, JINIS, 2018

05

NEOINTIMA PROCINE MODEL^{45*}

Key Outcome:
Earlier and thicker neointima growth

Published: Matsuda Y. et al. Analysis of neointima development in flow diverters using optical coherence tomography, JINIS, 2017.

06

BENCH TESTING^{44*}

Key Outcome:
Reduced delivery and resheathing forces

Medtronic Internal Study, TR-NV11991 Rev. B, Performance Evaluation of the Pipeline Flex Embolization Device with Shield Technology.

*Data is derived from referenced pre-clinical studies and bench testing and may not be representative of clinical performance.



INTRODUCTION

BACKGROUND

DESIGN

IN-SERVICE

CLINICAL

REFERENCES

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Shield Technology™



Treat with Ease*

*Data is derived from referenced pre-clinical studies and bench testing and may not be representative of clinical performance.

Overall Design

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Shield Technology™

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This hydrophilic property of Shield Technology™ leads to an improvement in deliverability and resheathing forces.^{44,52*}
The polarity of the phosphorylcholine creates a hydrophilic surface thus reducing friction in the presence of water, saline or blood.



The significant decrease in water contact angle from A (bare Co-Cr to resemble Pipeline Flex) to B (Co-Cr with Shield Technology™ to resemble Pipeline Flex with Shield Technology™) indicates hydrophilicity as a result of the surface modification.^{44*}

*Data is derived from referenced bench testing and may not be representative of clinical performance.



Overall Design

Braid Design

Delivery System Design

Shield Technology™

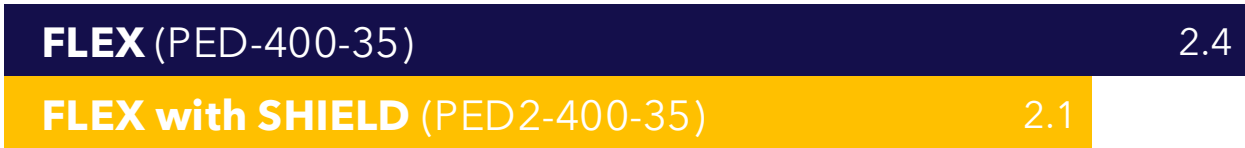
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Shield Technology™ improves the physician experience through improvements in delivery and resheathing force through tortuosity.^{44*}

Delivery Force (N)



Resheathing Force (N)

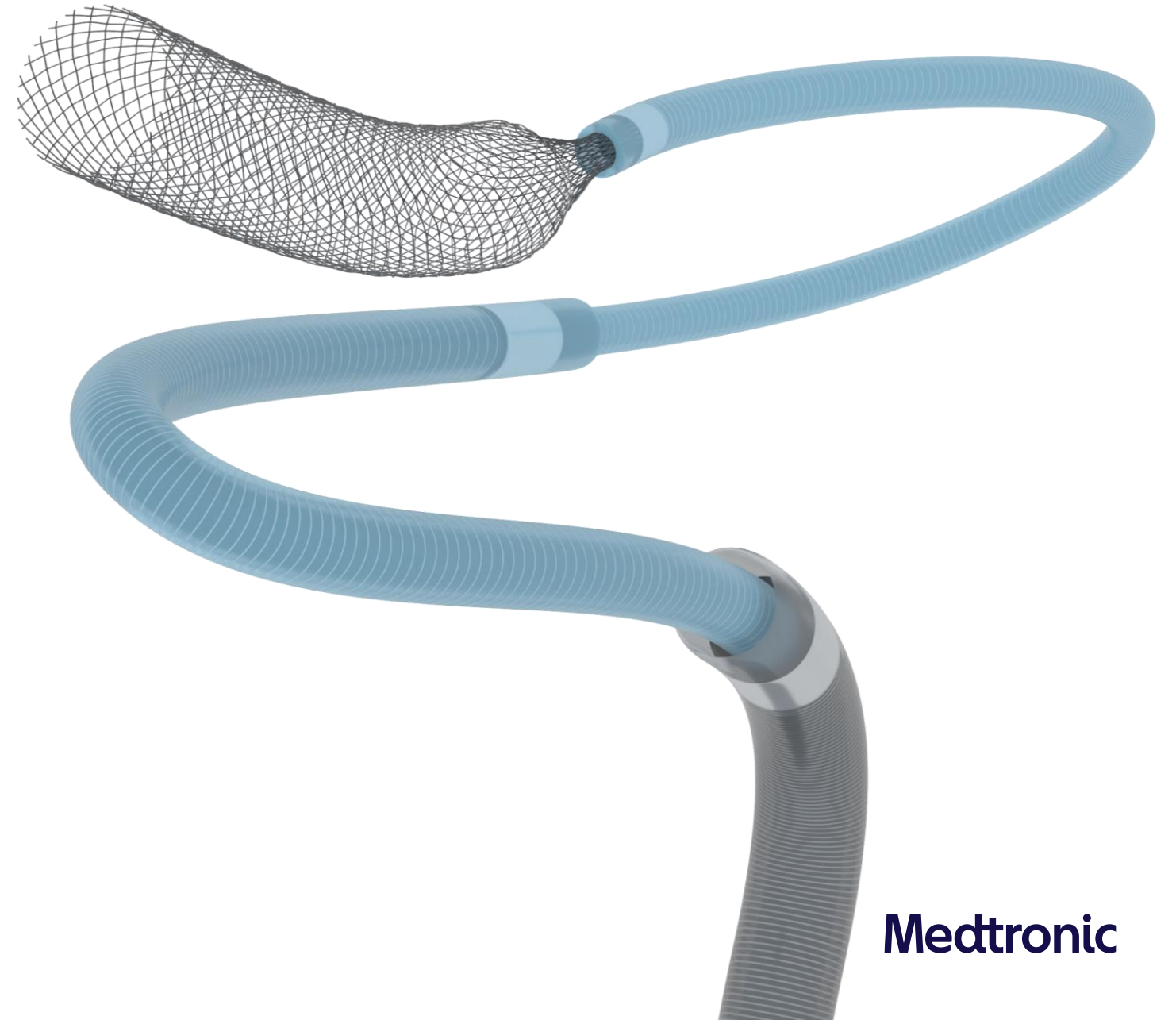


Bench model testing show that the modified surface of the implant exhibits hydrophilic properties, **decreasing delivery and resheathing forces by 20% and 12%, respectively.**^{44*}

*Data is derived from referenced bench testing and may not be representative of clinical performance.



Pipeline™ In-Service

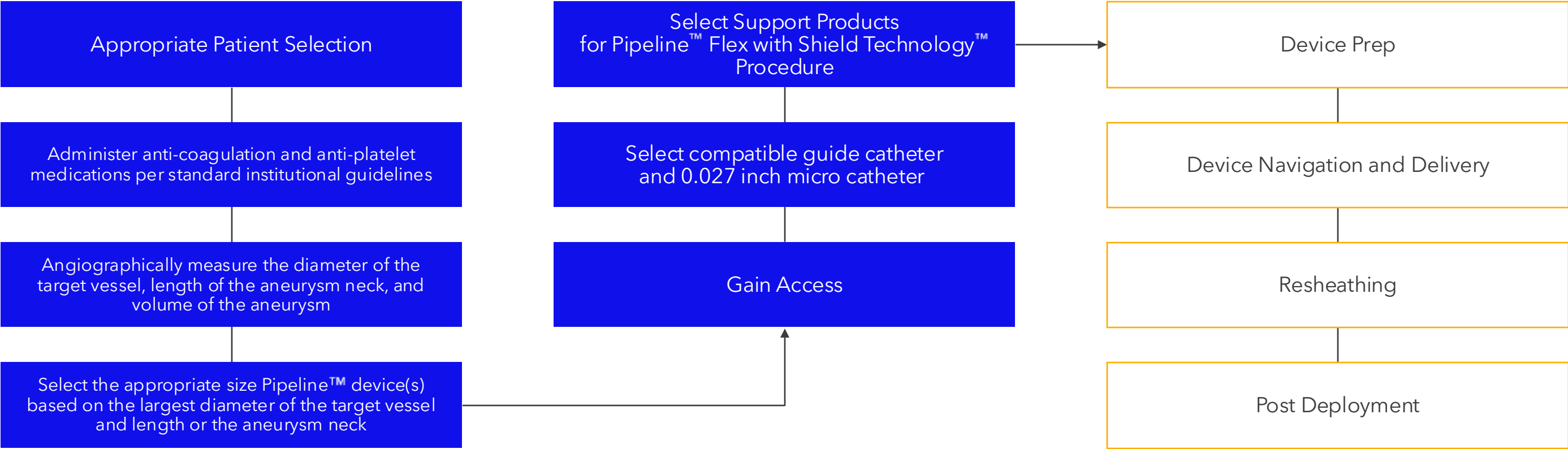


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Procedural Flow^{5,42}

■ Pre-procedure ■ Procedure



*For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵



Evolution of Access

Rist™ Radial Access System

Combine to Conquer

The Rist™ Radial Access System

features the first guide catheter designed for the unique demands of accessing the neurovasculature through the radial pathway.⁵⁹⁻⁶³





Evolution of Access

Navien™ Intracranial Support Catheter, Phenom™ 027 Catheter and Phenom™ Plus Catheter

Navien™ Intracranial Support Catheter

- The Navien™ intracranial support catheter has a nitinol coil and proprietary polymers to deliver thin wall construction that provides a large ID among 5 and 6 Fr catheters while delivering improved trackability and resistance to ovalization.⁶⁴⁻⁶⁶

Phenom™ 027 Catheter

- Composite design to maintain stability, and facilitate resheathing during flow diversion procedures^{43,55}
- The lumen of the catheter is strategically engineered to be durable for delivering advanced therapeutic devices with minimal friction while minimizing ovalization⁴³

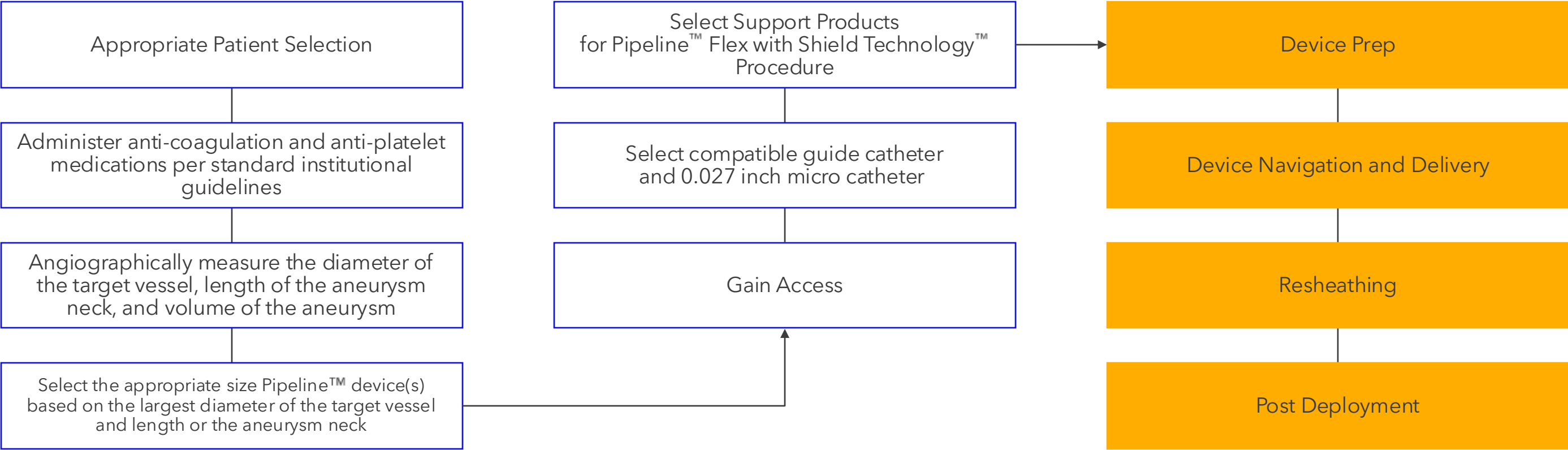
Phenom™ PLUS Catheter

- A robust support catheter with hydrophilic coating that is designed to facilitate accessing distal anatomy^{43,56}
- Low profile support system with 4.2F Distal OD⁴³



Procedural Flow^{5,42}

■ Pre-procedure ■ Procedure



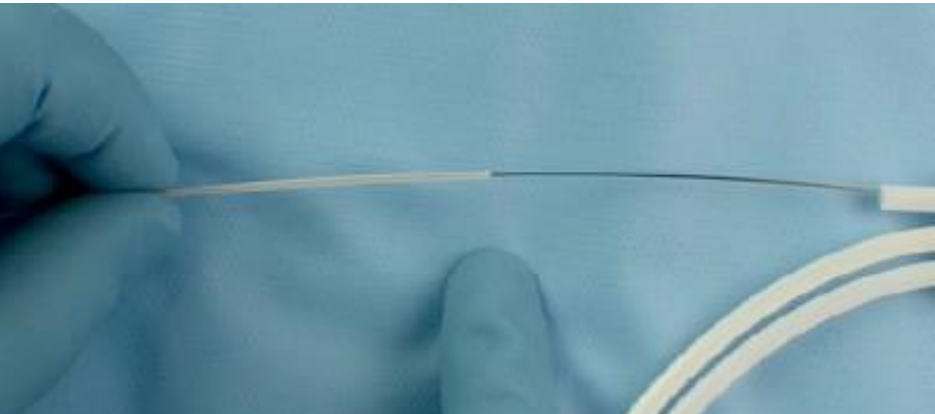
*For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵



Device Prep⁵



- ▶ Pull the distal end of the introducer sheath from the blue clip on the packaging hoop.



- ▶ Carefully remove system from packaging hoop until the delivery wire is exposed.

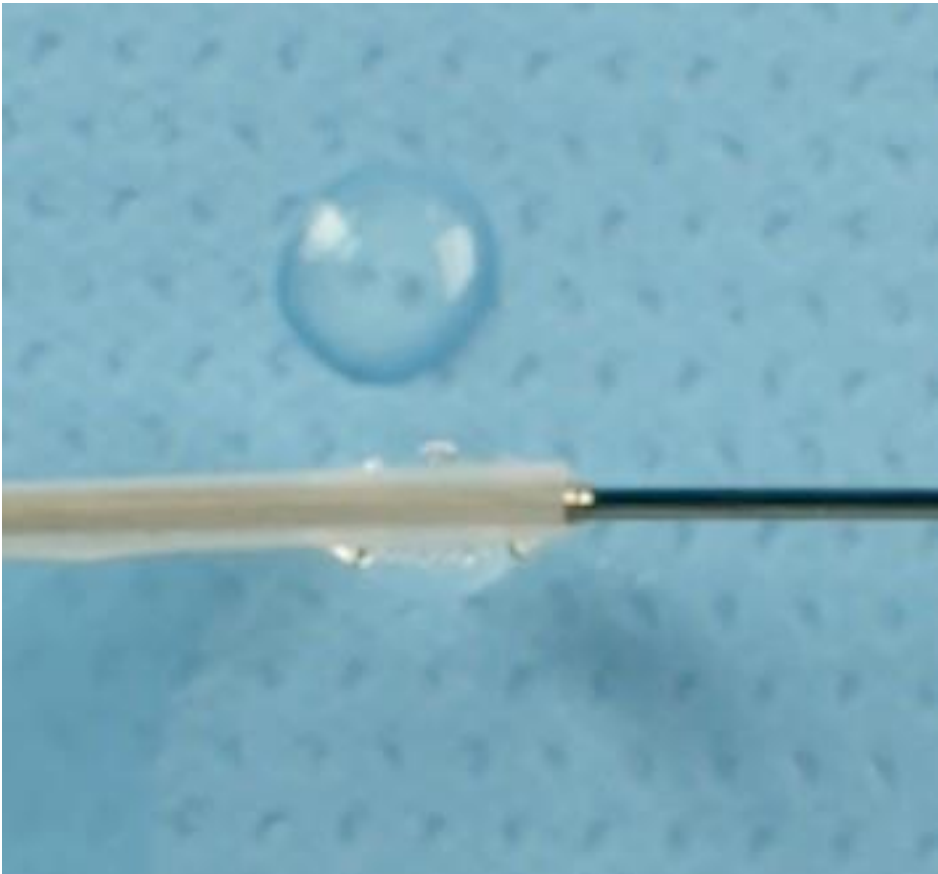
⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵

Image property of Medtronic





Loading the Device⁵



- ▶ Partially insert the introducer sheath into the rotating hemostatic valve (RHV) at the catheter hub and close the RHV.
- ▶ Using a flush pressure of 250 mmHg or greater, confirm back flush of the saline at the proximal end of the introducer sheath.
- ▶ Advance the introducer sheath into the RHV; visually confirm the tip of the sheath is deeply in the hub of the micro catheter.
- ▶ Secure introducer sheath to the hub by locking down the RHV tightly.
- ▶ Advance the proximal end of the delivery wire until it aligns with the proximal end of the introducer sheath. Remove the introducer sheath.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵

Image property of Medtronic

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Fluorosafe Visualization⁵



▶ Fluoroscopy is not required during the process of pushing the proximal end of the delivery wire to the proximal end of the introducer sheath.



▶ The delivery wire has a Fluorosafe Marker no further than 125 cm from the distal end of the device.

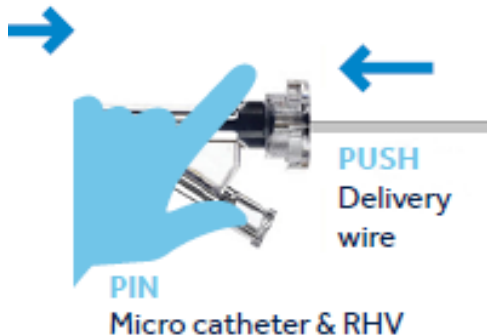
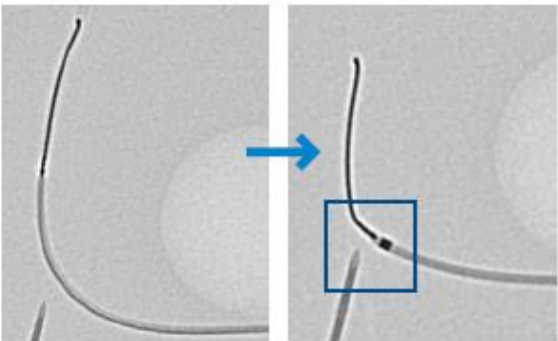


Caution: The Fluorosafe Marker is only compatible with microcatheters with a minimum length of 135 cm.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵



Device Navigation⁵

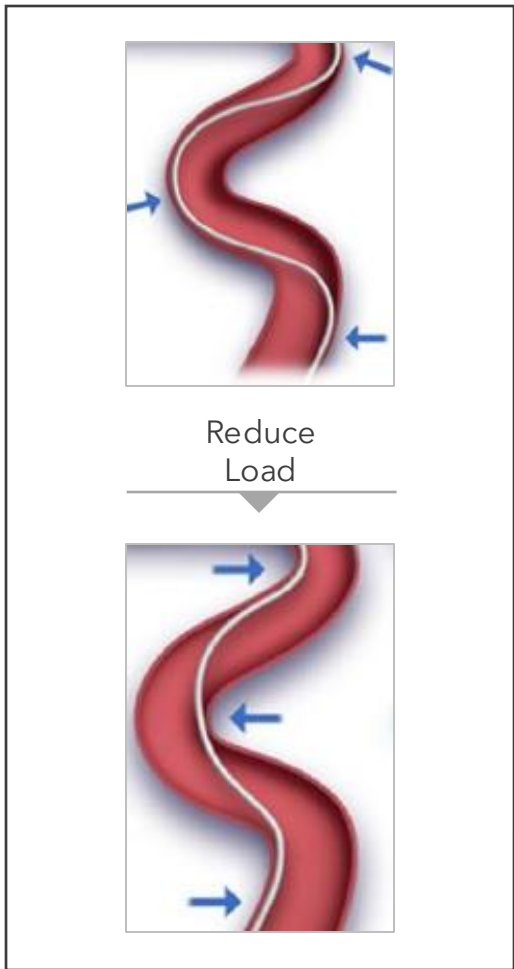


- ▶ Advance the Pipeline™ device into the micro catheter by pushing the delivery wire.
- ▶ Once the tip of delivery system and micro catheter are aligned, verify that the Pipeline™ device is in the desired location. The Pipeline™ device should be placed at least 3mm past the distal edge of the aneurysm.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵



Tips & Techniques^{5*}



When advancing the Pipeline™ system through micro catheter, continually reduce the load in system.

Reduce Load by:

- 01** Slowly pulling back on system (micro catheter and delivery wire together).
- 02** Continue unloading until advancement of Pipeline™ device braid begins inside of micro catheter. Ensure distal tip does not lose position.
- 03** Continue to push wire while maintaining reduced load in micro catheter.

CAUTION: If high force or excessive friction is encountered during delivery, discontinue delivery of the device and identify the cause of the resistance. Remove device and micro catheter simultaneously. Advancement of the Pipeline™ device against resistance may result in damage or patient injury.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵
^{*}Based on physician feedback



Initial Deployment⁵



Begin to deliver the Pipeline™ device using a combination of unsheathing and pushing the delivery wire simultaneously.

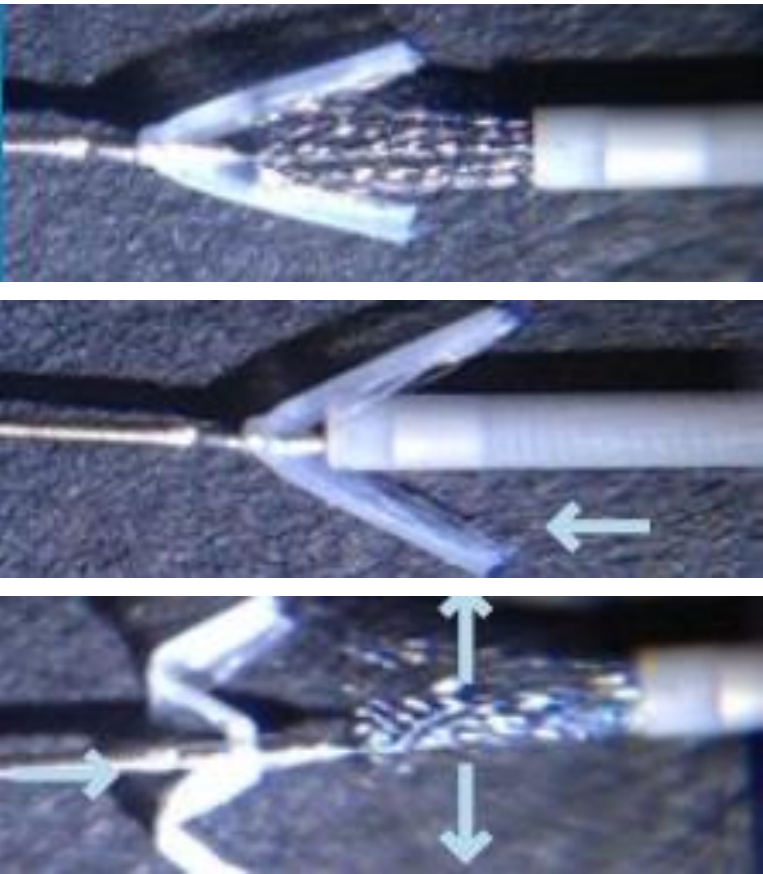


The braid will remain constrained until PTFE sleeves are outside micro catheter.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵



Distal Braid Opening⁵



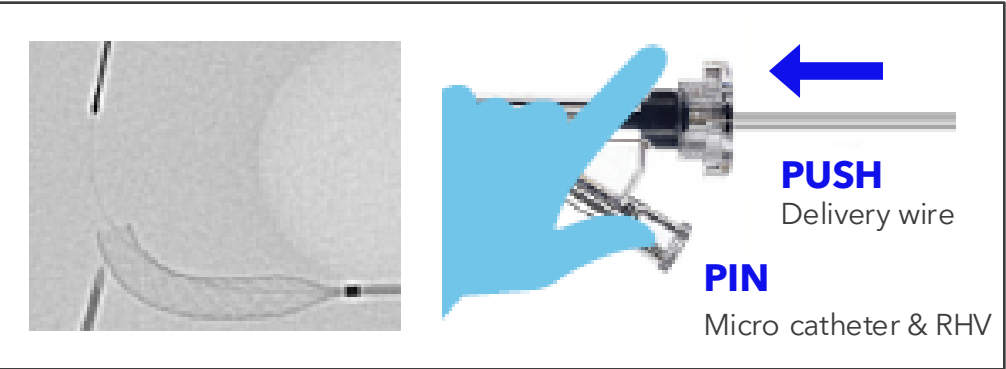
If PTFE sleeves are not separated from braid, resheath distal braid to push PTFE sleeves forward. Afterwards, distal braid is free to expand.*

Tip: If the braid is slow in opening, advance more braid (push more wire), then unload braid off vessel wall.*

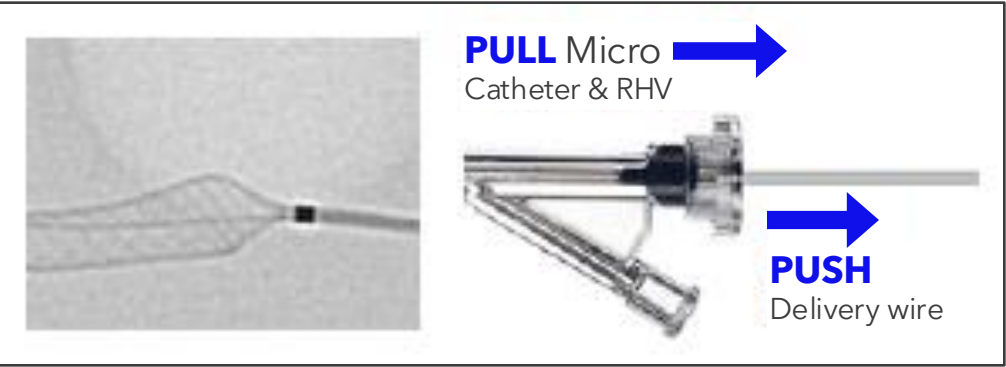
*For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵
*Based on physician feedback



Distal Braid Opening⁵



▶ Continue deploying the braid by pushing the delivery wire while pinning the micro catheter.



▶ If too much load builds up on the braid, adjust system by pulling micro catheter and delivery wire together.*

*For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵
*Based on physician feedback



Tips & Techniques^{5*}

Device Prep

Device Navigation
and Delivery

Resheathing

Post Deployment

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Unloading the System

Decreasing forward pressure on the system by pulling back on the micro catheter and delivery wire together

Application

During deployment to pull micro catheter from outer curvature to inner curvature



Indicates
deployment angle



Indicates catheter
position in vessel

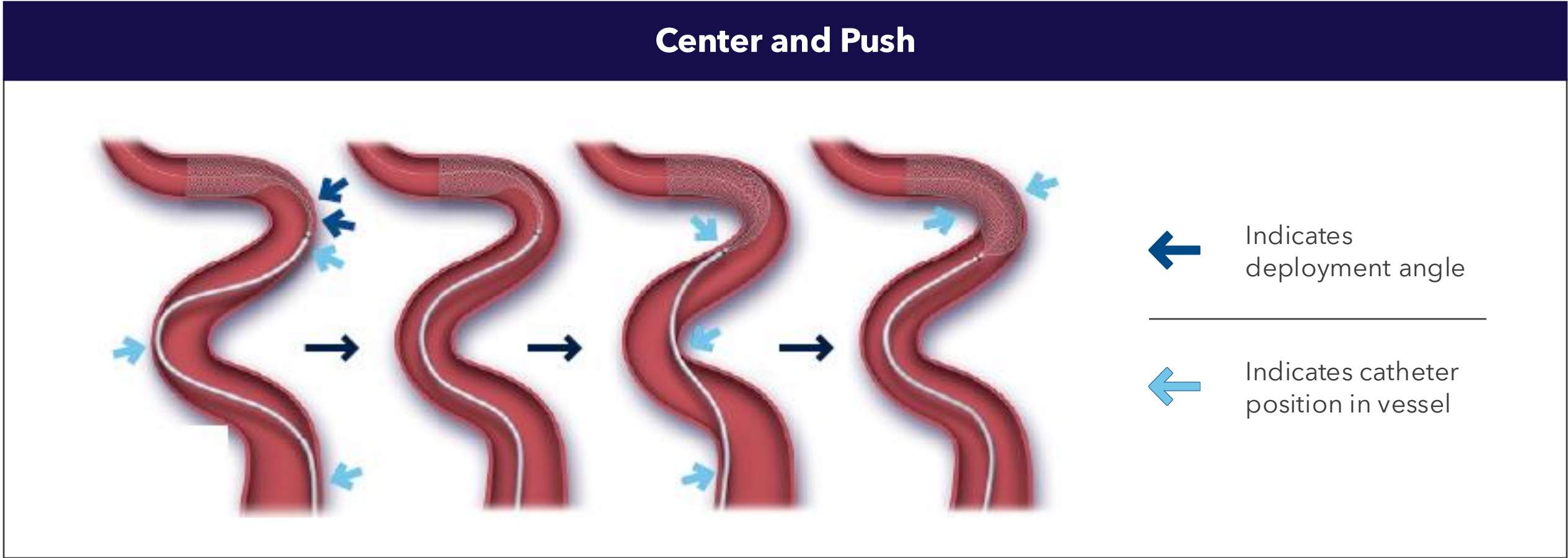
⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵
^{*}Based on physician feedback

Image property of Medtronic

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Tips & Techniques^{5*}



⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵
^{*}Based on physician feedback

Image property of Medtronic

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Loading Technique and Deployment Angles^{5*}

Device Prep



✓ Appropriate amount of forward push on delivery wire.

Device Navigation and Delivery



✗ Too much unsheathing of the micro catheter may lead to temporary stretching of the device.

Resheathing



✗ Too much forward push on the micro catheter may lead to temporary compression of the device.

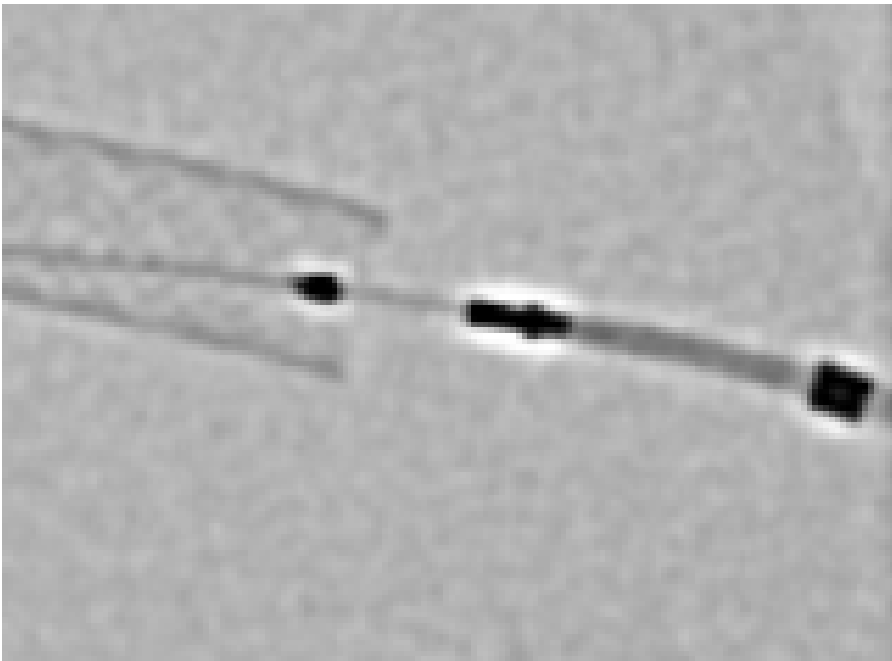
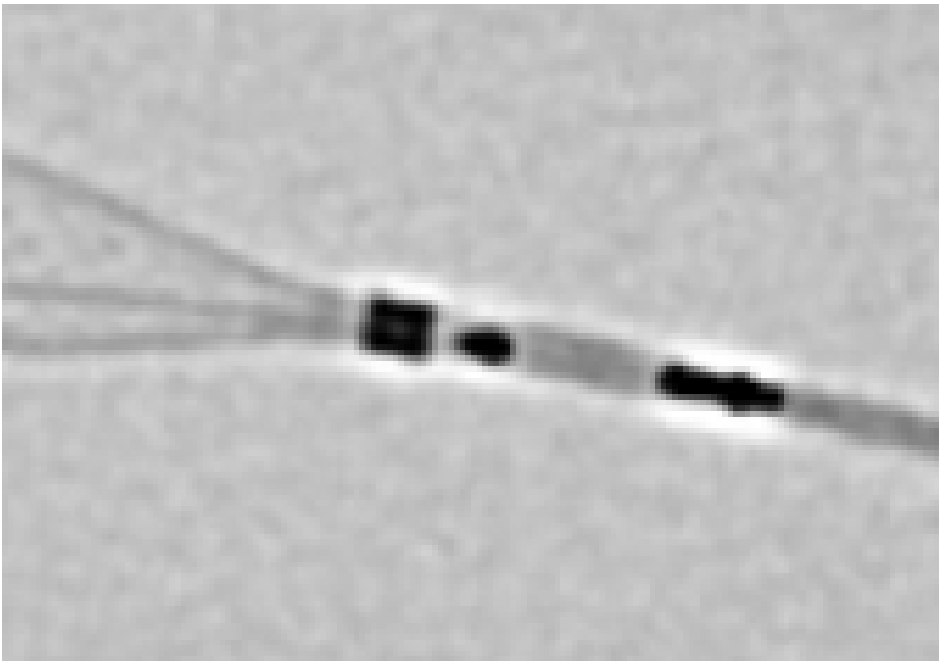
Post Deployment

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⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵
^{*}Based on physician feedback



Proximal Deployment⁵



If resheathing is not desired, deploy the proximal end of the device by unsheathing the braid from the micro catheter

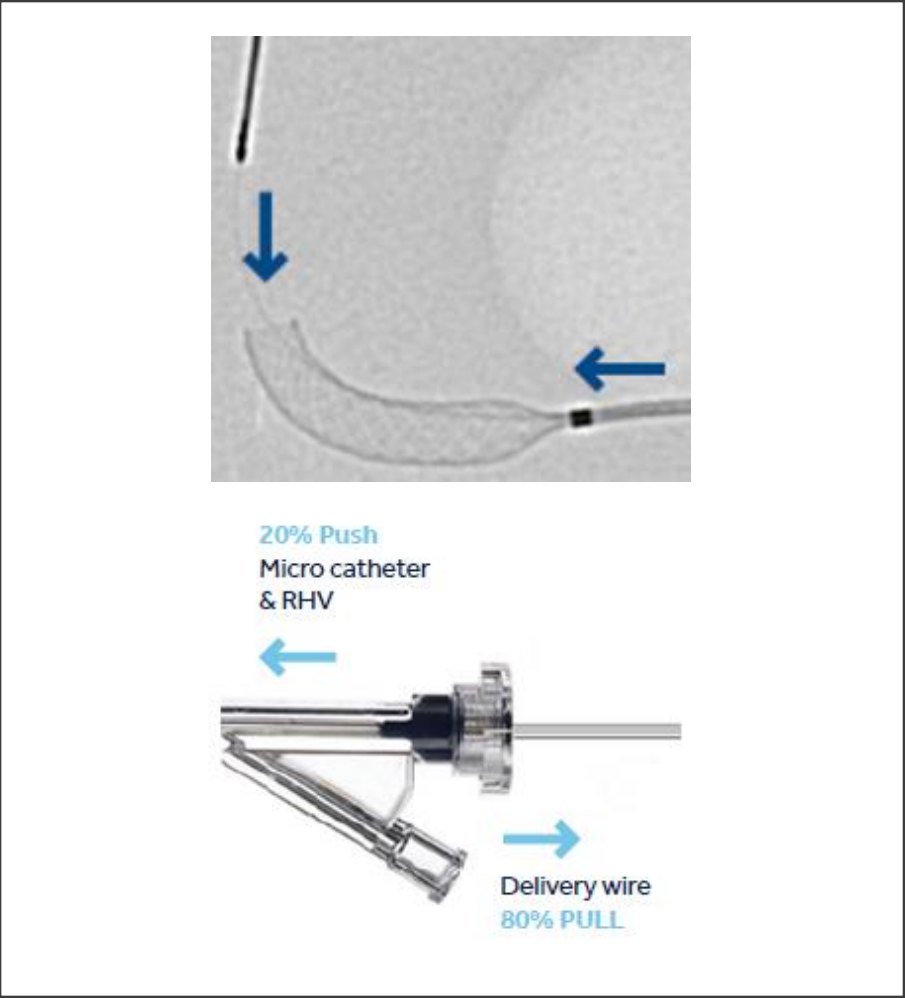
⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵

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Resheathing⁵

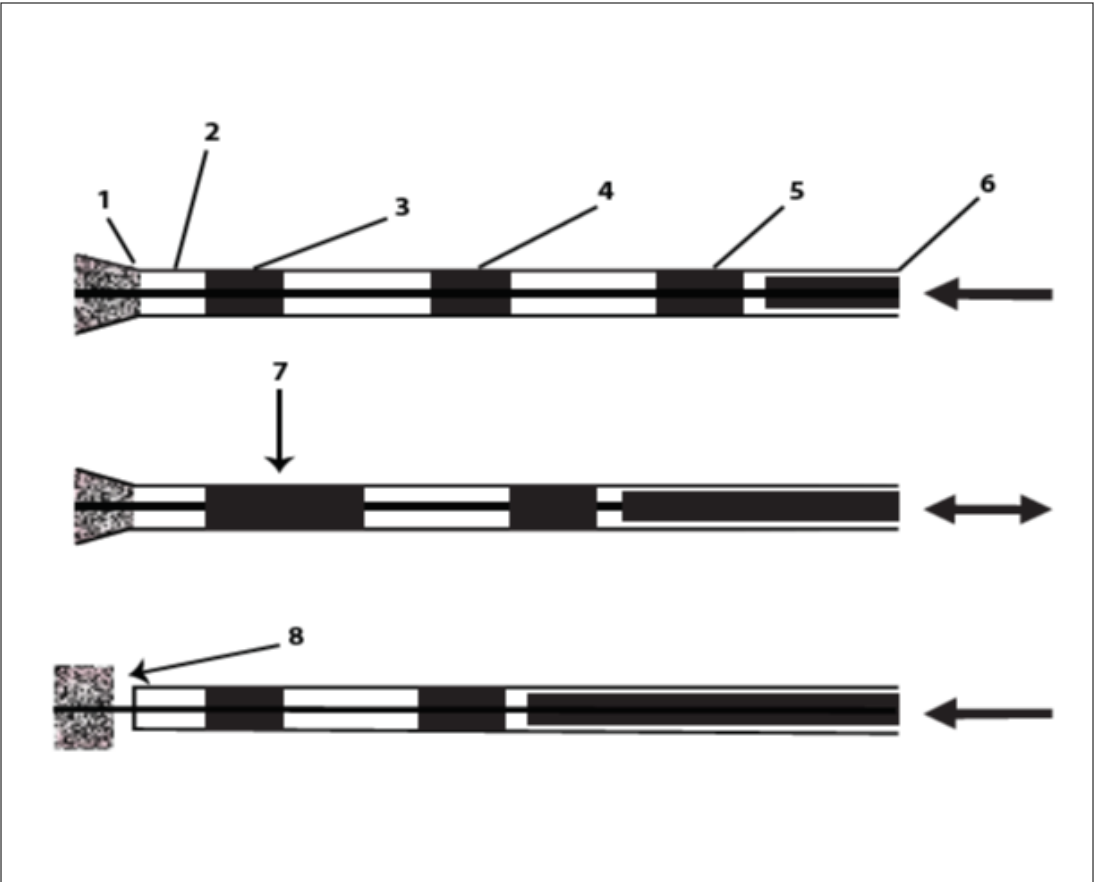


- ▶ During deployment of the Pipeline™ device, resheathing can be performed by advancing the micro catheter while pulling delivery wire.
- ▶ The Pipeline™ device can be resheathed until the resheathing marker has reached the distal marker of the micro catheter.
- ▶ The Pipeline™ device is fully resheathed when the distal marker is retracted completely inside the micro catheter.
- ▶ The system is designed to allow for two full cycles of resheathing of the Pipeline™ device.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵



Resheathing schematic as seen under fluoroscopy (image not to scale)⁵



- 01 Proximal End of Device
- 02 Micro Catheter
- 03 Micro Catheter Distal Marker
- 04 Resheathing Marker
- 05 Proximal Bumper
- 06 Delivery Wire
- 07 Resheathing Limit
- 08 Device Detached

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use



Resheathing⁵



Begin resheathing by removing energy from the system. Bring micro catheter off of the greater curvature of the vessel.

Put slight forward pressure on micro catheter and mostly pull delivery wire.

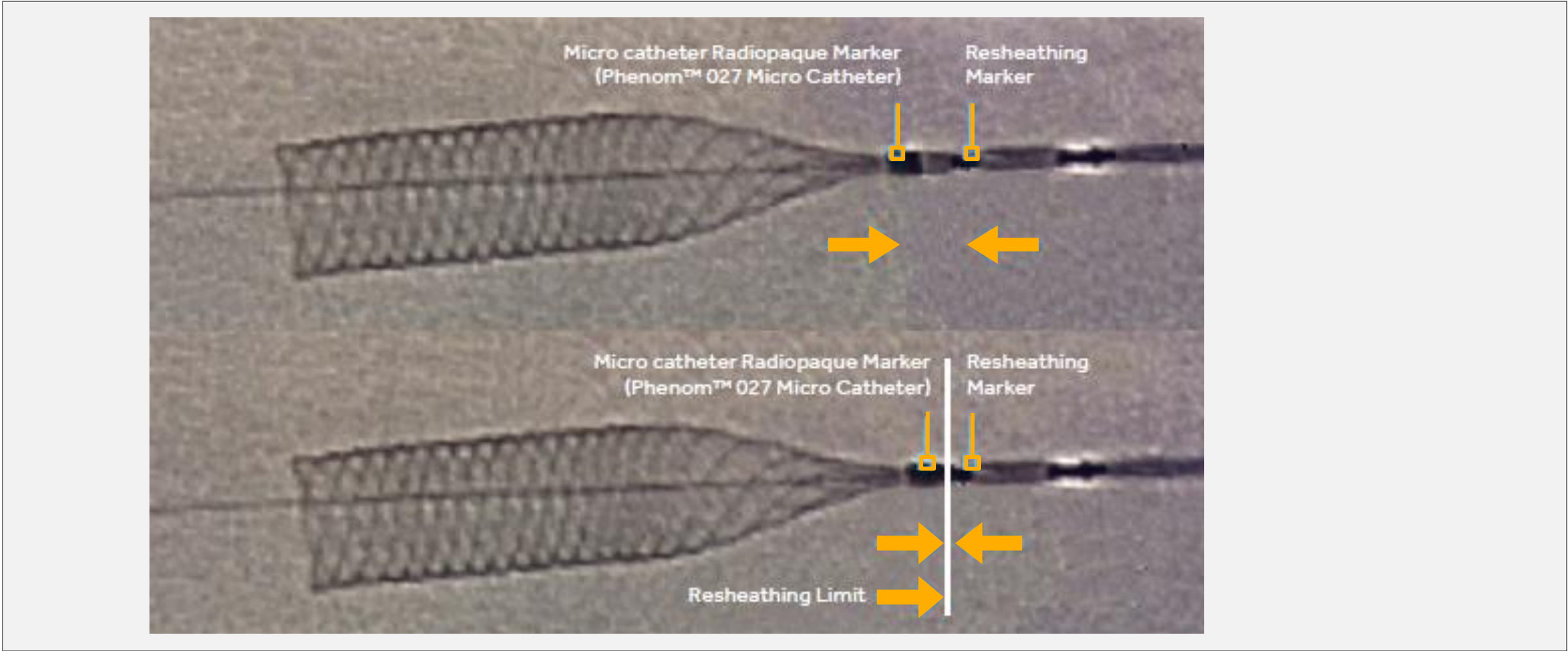
The point of no return is the distal end of the resheathing marker.

Take note of braid behavior and load on the system while attempting to resheath.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use



Resheathing⁵



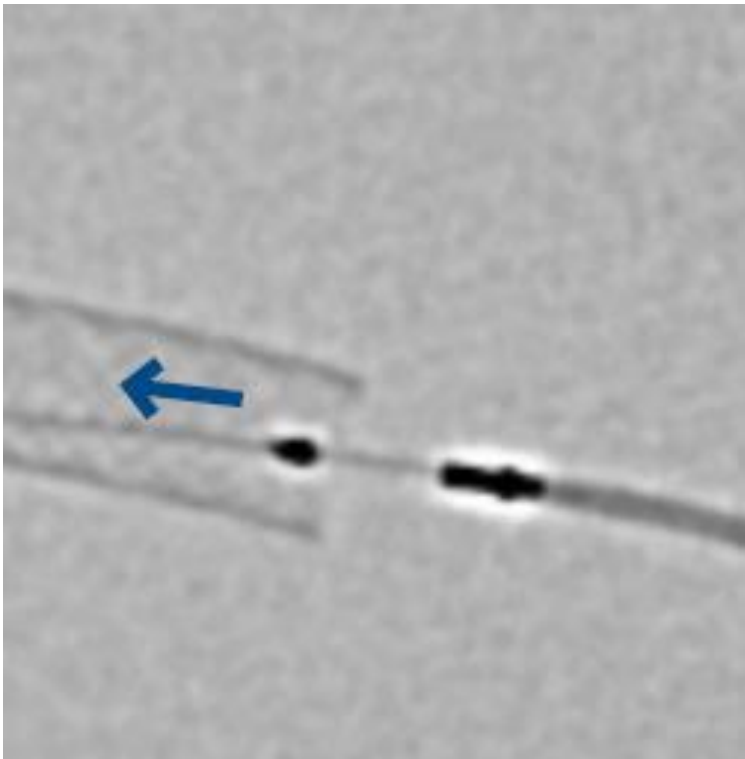
*For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵

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Post Deployment⁵



▶ After the Pipeline™ device is deployed, advance the micro catheter through the device making sure not to dislodge the braid.

▶ When the micro catheter tip is distal to the braid, retract the delivery wire into the micro catheter tip.

Caution: If the catheter cannot be advanced through the braid, carefully remove the delivery wire through the implant construct.

Caution: If the delivery wire cannot be retracted into the micro catheter, carefully remove the core wire and micro catheter simultaneously.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use⁵

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Post Deployment⁵

Device Prep
Device Navigation and Delivery
Resheathing
Post Deployment

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Carefully inspect the deployed Pipeline™ under fluoroscopy to confirm that it is completely apposed to the vessel wall and not kinked.

Note: If the device is not fully apposed or is kinked, consider using a balloon catheter, micro catheter, or guidewire to fully open the device.

⁵For additional information, consult the Pipeline™ Flex with Shield Technology™ Instructions for Use



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Indications for Use: The Pipeline™ Flex Embolization Device with Shield Technology™ is indicated for the endovascular treatment of adults (22 years of age or older) with large or giant wide-necked intracranial aneurysms (IAs) in the internal carotid artery from the petrous to the superior hypophyseal segments. The Pipeline™ Flex Embolization Device with Shield Technology™ is also indicated for use in the internal carotid artery up to the terminus for the endovascular treatment of adults (22 years of age or older) with small and medium wide-necked (neck width ≥ 4 mm or dome-to-neck ratio < 2) saccular or fusiform intracranial aneurysm (IAs) arising from a parent vessel with a diameter ≥ 2.0 mm and ≤ 5.0 mm.